MIL-C-14005A
2 August 1967
SUPERSEDING
MIL-C-14005(CmlC)
21 March 1955

MILITARY SPECIFICATION

CUPRIC CARBONATE, BASIC, ANALYZED REAGENT

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers reagent grade basic cupric carbonate.

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-621	-	Boxes, Wood, Nailed and Lock-corner.
PPP-B-636	-	Box, Fiberboard.
PPP-C-301	-	Chemicals, Dry and Paste, Packaging and Packing of.
PPP-F-320	-	Fiberboard; Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes.

MILITARY

MIL-P-3938	-	Pallet, Material Handling, Hardwood, Stringer Construction, 4- Way (Partial).

FSC 6810

Source: https://assist.dla.mil -- Downloaded: 202610-18T02:45Z
Check the source to verify that this is the current version before use.

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	-	Marking for Shipment and Storage.
MIL-STD-147	-	Palletized and Containerized Unit Loads, 40" x 48", 4-Way (Partial) Pallet Skids, Runners or Pallet-type Base.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

UNIFORM CLASSIFICATION COMMITTEE

Uniform Freight Classification

(Application for copies of these ratings, rules and regulations should be addressed to Uniform Classification Committee, 202 Union Station, 516 West Jackson Boulevard, Chicago, Illinois 60606.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM Standards

D1193-66	-	Reagent Water.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.

Source: https://assist.dla.mil -- Downloaded: 202640-18T02:45Z
Check the source to verify that this is the currentversion before use.

3. REQUIREMENTS

3.1 Appearance. Basic cupric carbonate shall be a fine green powder.

3.2 Chemical characteristics. Basic cupric carbonate shall conform to the chemical characteristics of table I when tested as specified therein.

Table I. Chemical characteristics

Characteristic	Percent by weight		Test paragraph
	Minimum	Maximum	
Assay (as Cu)	53.0	---	4.2.4.1
Insoluble matter in HCl	----	0.020	4.2.4.2
Chloride (Cl)	----	0.01	4.2.4.3
Nitrate (NO_3)	----	0.05	4.2.4.4
Sulfate (SO_4)	----	0.01	4.2.4.5
Substances not precipitated by H_2S (as sulfates)	----	0.50	4.2.4.6
Iron (Fe)	----	0.01	4.2.4.7

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.

4.1.1 Supplier's responsibility. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.2 Objective evidence. The supplier shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1 and section 5 for which specific inspection has not been provided in this specification have been satisfied.

Source: https://assist.dla.mil -- Downloaded: 202610-18T02:45Z
Check the source to verify that this is the current version before use.

4.2 Regular production.

4.2.1 Lotting. A lot shall consist of the cupric carbonate produced by one manufacturer, at one plant, from the same materials, and under essentially the same manufacturing conditions provided the operation is continuous. In the event the process is a batch operation, each batch shall constitute a lot (see 6.3).

4.2.2 Sampling for test. Sampling shall be conducted in accordance with MIL-STD-105, level S-2. Approximately equal portions of cupric carbonate shall be taken from each container in the sample, composited, and placed in a suitable clean, dry container labeled to identify the lot and containers from which it was taken. The total composited sample shall be no less than 1/4 pound.

4.2.3 Inspection. The composited sample of cupric carbonate shall be tested as specified in 4.2.4. Each test shall be conducted in duplicate analysis. Failure of either analysis of any test shall be cause for rejection.

4.2.4 Tests. Water in accordance with ASTM D1193 and reagent grade chemicals shall be used throughout the tests. Where applicable, blank determinations shall be run and corrections applied where significant. Tests shall be conducted as follows:

4.2.4.1 Assay (as Cu). Heat approximately 10 grams (g) of the specimen in an oven at 105° + 5°C for 1 hour and then allow to cool in a desiccator. Weigh to the nearest 0.1 milligram (mg) approximately 0.4 g of the dried specimen into a glass-stoppered flask and dissolve, with heat, in 20 milliliters (ml) of water and 5 ml of phosphoric acid. Cool the solution and add 80 ml of water and 3 g of potassium iodide. Titrate the liberated iodine with standardized 0.1N sodium thiosulfate solution, adding starch solution toward the end as indicator. Calculate the percent copper as follows:

$$\text{Percent copper} = \frac{6.36AB}{W}$$

where: A = Normality of sodium thiosulfate solution,
B = Milliliters of sodium thiosulfate solution used, and
W = Weight of specimen in grams.

4.2.4.2 Insoluble matter in hydrochloric acid. Weigh to nearest 0.1 mg approximately 10 g of the specimen into a 250-ml beaker. Add 100 ml of water and 10 ml of hydrochloric acid and heat until the specimen is dissolved. Filter

Source: https://assist.dla.mil -- Downloaded: 202640-18T02:45Z
Check the source to verify that this is the current version before use.

any undissolved residue in a tared filter crucible, wash thoroughly with hot water, dry to constant weight at 105° ± 5°C, cool, and weigh. Calculate the percent insoluble matter as follows:

$$\text{Percent insoluble matter} = \frac{100\,(A-B)}{C}$$

where: A = Weight of crucible and residue in grams,
B = Weight of crucible in grams, and
C = Weight of specimen in grams.

4.2.4.3 Chloride. Weigh 0.300 g of the specimen and dissolve in 20 ml of water and 2 ml of nitric acid. Filter, if necessary. Divide the filtrate into two equal portions. To one portion add 1 ml of 0.1N silver nitrate solution, bring to a boil, let stand for 10 minutes, cool, filter until clear, and use for the control. To the other portion add 1 ml of 0.1N silver nitrate solution. Any turbidity which results shall be no greater than that produced by adding 0.015 mg of chloride to the control and adjusting both portions to the same volume.

4.2.4.4 Nitrate.

(a) Indigo carmine solution. Dissolve 0.10 g of indigo carmine, previously dried for 2 hours at 105° ± 5°C, in a mixture of 10 ml of sulfuric acid and 80 ml of water and dilute with water to make 100 ml.

(b) Procedure. Weigh 1.000 g of the specimen and dissolve in 15 ml of water and 2 ml of hydrochloric acid. Add the solution, with stirring, to 20 ml of a 10-percent sodium hydroxide solution and digest on a steam bath for 15 minutes. Cool the solution, dilute to 40 ml with water, and filter. To 4 ml of the filtrate, add 0.05 ml of the indigo carmine solution and 10 ml of sulfuric acid. The blue color shall persist for no less than 5 minutes.

4.2.4.5 Sulfate.

(a) Potassium sulfate solution. Dissolve 0.1814 g of potassium sulfate in water to make 1 liter. One ml contains 0.1 mg of sulfate.

(b) Procedure. Weigh 2.000 g of the specimen and dissolve in 20 ml of water and 3 ml of hydrochloric acid. Dilute the solution with water to make 100 ml and filter. To 50.0 ml of the filtrate add 3 ml of 12-percent barium chloride solution. Any turbidity which results shall be no greater than that produced when 3 ml of 12-percent barium chloride solution and 1.0 ml of the potassium sulfate solution are added to 49.0 ml of water.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T02:45Z
Check the source to verify that this is the current version before use.

4.2.4.6 Substances not precipitated by hydrogen sulfide. Weigh to the nearest 0.1 mg approximately 1 g of the specimen and dissolve in 100 ml of water and 2 ml of sulfuric acid. Heat the solution to 70° ± 5°C and precipitate the copper with hydrogen sulfide. Allow the precipitate to settle and filter without washing. Evaporate 50 ml of the filtrate to dryness and ignite to constant weight. (Save the residue for 4.2.4.7.) Calculate the percent substances not precipitated by hydrogen sulfide as follows:

$$\text{Percent substances not precipitated by hydrogen sulfide} = \frac{200\ A}{W}$$

where: A = Weight of residue in grams and
W = Weight of specimen in grams.

4.2.4.7 Iron.

(a) Ammonium thiocyanate solution. Dissolve 25 g of ammonium thiocyanate in water to make 100 ml.

(b) Procedure. To the residue from 4.2.4.6, add 2 ml of hydrochloric acid, 2 ml of water, and 0.5 ml of nitric acid, then slowly evaporate to dryness on a steam bath. Take up the residue in 1 ml of hydrochloric acid and 10 ml of warm water, and dilute with water to 50 ml. To 10 ml of this solution, add 2 ml of hydrochloric acid and dilute to 50 ml. Add approximately 30 mg of ammonium persulfate crystals and 3 ml of the ammonium thiocyanate solution, and mix. Any red color produced shall be no greater than that in a control made with 0.01 mg of iron, using the same quantities of reagents and in the same final volume as the specimen.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A. One pound ± 0.05 ounce of cupric carbonate shall be packaged in a screw cap bottle in accordance with the requirements for level A packaging of chemicals in type I, class 1 unit containers utilizing inclosure b (bottles in metal cans) as specified in PPP-C-301.

5.1.1.1 Intermediate packaging. Twelve bottles of cupric carbonate, packaged as specified in 5.1.1, shall be intermediately packaged in an upright position in a fiberboard box in a three bottle by four bottle pattern. The approximate inside dimensions of the box shall be 23 inches by 17-1/2 inches by 8 inches. Fiberboard pads shall be inserted where needed to assure a tight pack. The box shall conform

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T02:45Z
Check the source to verify that this is the current version before use.

to style RSC, grade W6c, for a type 2 average load of PPP-B-636. Pads shall conform to type CF, class domestic, variety SW, grade 125 of PPP-F-320. Boxes shall be closed in accordance with requirements for closure of class weather resistant boxes as specified in the appendix to PPP-B-636.

5.1.2 Level C. Quantities of cupric carbonate as specified (see 6.2) shall be packaged to provide adequate protection against corrosion, deterioration and damage from the supply source to the first receiving activity for immediate use.

5.2 Packing. Cupric carbonate shall be packed level A, B, or C as specified (see 6.2).

5.2.1 Level A. Four boxes of cupric carbonate, intermediately packaged as specified in 5.1.1.1, shall be packed for shipment in a nailed wood box conforming to class 2, style 4-1/2, for a type 2 average load of PPP-B-621. The box shall be closed and strapped in accordance with the applicable requirements as specified in the appendix to PPP-B-621.

5.2.2 Level B. Cupric carbonate shall be packed as specified in 5.2.1 except that a fiberboard box shall be used instead of the nailed wood box. The fiberboard box shall conform to style RSC, grade V3c of PPP-B-636, and shall be closed by taping in accordance with the requirements for taping style RSC boxes to be shipped as exterior containers and then reinforced by tape banding as specified in the appendix to PPP-B-636.

5.2.3 Level C. Cupric carbonate, packaged as specified in 5.1.2, shall be packed to insure carrier acceptance and safe delivery to the first domestic destination. Containers shall be in compliance with Uniform Freight Classification rules or the regulations of other common carriers applicable to the mode of transportation.

5.3 Marking. In addition to the special marking for unit containers specified in 5.3.1 and any special marking required by the contract or order, unit and intermediate packages and shipping containers shall be marked in accordance with MIL-STD-129 (see 6.2).

5.3.1 Special marking. Each unit container shall be durably and legibly labeled to show the following special information:

(a) Empirical formula.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T02:45Z
Check the source to verify that this is the current version before use.

(b) Formula weight.
(c) Manufacturer's lot number.
(d) Assay.
(e) Analysis or maximum limits of impurities.

5.4 Palletization. When specified in the contract or order, shipping containers shall be palletized in accordance with the applicable requirements of MIL-STD-147 utilizing the 40 by 48 inch, double-wing pallet of MIL-P-3938 (see 6.2).

6. NOTES

6.1 Intended use. Cupric carbonate is intended for use as a general laboratory reagent.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Level of packaging and packing required.
(c) Quantity required for level C packaging.
(d) Special marking other than 5.3.1, if required.
(e) Palletization, if required.

6.3 Batch. A batch is defined as that quantity of material which has been manufactured by some unit chemical process and subjected to some physical mixing operation intended to make the final product substantially uniform.

6.4 Significant places. For the purpose of determining conformance with this specification, an observed or calculated value shall be rounded off "to the nearest unit" in the last right-hand place of figures used in expressing the limiting value, in accordance with the rounding-off method of the Recommended Practices for Designating Significant Places in Specified Limiting Values (ASTM E29).

Custodians:

Army - MU
Air Force - 68

Preparing activity:

Army - MU(EA)

Review activities:

Army - MU(FA)
DSA - GS

Project No. 6810-0396

User activities:

Army - MI, SM

Coordination Method Code: C

Source: https://assist.dla.mil -- Downloaded: 202610-18T02:45Z
Check the source to verify that this is the current version before use.

TO DETACH FORM - CUT ON THIS LINE

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 56 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

Source: https://assist.dla.mil -- Downloaded: 202610-18T02:45Z
Check the source to verify that this is the current version before use.

FOLD

POSTAGE AND FEES PAID

OFFICIAL BUSINESS

Commanding Officer
US Army Edgewood Arsenal
ATTN: SMUEA-QAWS
Edgewood Arsenal, Maryland 21010

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T02:45Z
Check the source to verify that this is the current version before use.